Srushti Matkar B25ET1103- MDM 5

Mini- project

OUTPUT

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DATA PREVIEW

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student_id gender year main_device screen_hours breaks_per_day eye_discomfort concentration_difficulty breaks_per_hour eye_label conc_label screen_group break_group eye_group conc_group

1 Male 4 Both 7.4 4 3 2 0.54 Sometimes Rarely High (>7 h) Moderate (4-6) Moderate (3) Low (1-2)

2 Male 4 Laptop 4.7 9 3 2 1.91 Sometimes Rarely Medium (4-7 h) Frequent (7+) Moderate (3) Low (1-2)

3 Male 4 Both 8.3 7 1 2 0.84 Never Rarely High (>7 h) Frequent (7+) Low (1-2) Low (1-2)

4 Male 1 Both 8.7 3 3 1 0.34 Sometimes Never High (>7 h) Few (0-3) Moderate (3) Low (1-2)

5 Male 1 Both 2.9 9 1 1 3.10 Never Never Low (<=4 h) Frequent (7+) Low (1-2) Low (1-2)

6 Male 1 Both 4.2 3 2 2 0.71 Rarely Rarely Medium (4-7 h) Few (0-3) Low (1-2) Low (1-2)

7 Male 4 Both 7.1 8 2 3 1.13 Rarely Sometimes High (>7 h) Frequent (7+) Low (1-2) Moderate (3)

8 Male 4 Both 6.2 3 2 3 0.48 Rarely Sometimes Medium (4-7 h) Few (0-3) Low (1-2) Moderate (3)

9 Male 4 Both 6.8 7 2 2 1.03 Rarely Rarely Medium (4-7 h) Frequent (7+) Low (1-2) Low (1-2)

10 Male 1 Laptop 5.1 4 2 2 0.78 Rarely Rarely Medium (4-7 h) Moderate (4-6) Low (1-2) Low (1-2)

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1. DESCRIPTIVE STATISTICS (mean, median, std)

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	count	mean	median	std	min	max
screen_hours	300.0	6.72	6.60	1.86	2.0	12.6
breaks_per_day	300.0	5.02	5.00	2.22	0.0	13.0
breaks_per_hour	300.0	0.82	0.74	0.47	0.0	3.1
eye_discomfort	300.0	2.35	2.00	0.91	1.0	5.0
concentration_difficulty	300.0	2.08	2.00	0.88	1.0	5.0

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FREQUENCY DISTRIBUTIONS

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Eye discomfort

eye_label	Count	Percent
Never	55	18.3
Rarely	119	39.7
Sometimes	93	31.0
Often	32	10.7
Always	1	0.3

Concentration difficulty

conc_label	Count	Percent
Never	87	29.0
Rarely	116	38.7
Sometimes	86	28.7
Often	8	2.7
Always	3	1.0

Screen time group

	Count	Percent

screen_group

Low (<=4 h) 19 6.3

Medium (4-7 h) 150 50.0

High (>7 h) 131 43.7

Break group

Count Percent

break_group

Few (0-3) 83 27.7

Moderate (4-6) 142 47.3

Frequent (7+) 75 25.0

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MEAN / MEDIAN SYMPTOM SCORE BY BREAK GROUP

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eye_discomfort concentration_difficulty

mean median mean median

break_group

Few (0-3) 2.51 3.0 2.43 2.0

Moderate (4-6) 2.42 2.0 2.13 2.0

Frequent (7+) 2.04 2.0 1.59 1.0

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2. CORRELATION (Pearson and Spearman)

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Variable 1 Variable 2 Pearson r p (Pearson) Spearman rho p (Spearman)

screen_hours eye_discomfort 0.222 1.03e-04 0.211 2.37e-04

breaks_per_day eye_discomfort -0.203 4.09e-04 -0.206 3.22e-04

screen_hours concentration_difficulty 0.191 8.79e-04 0.185 1.31e-03

breaks_per_day concentration_difficulty -0.343 1.01e-09 -0.342 1.17e-09

eye_discomfort concentration_difficulty 0.274 1.38e-06 0.279 9.48e-07

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3. CHI-SQUARE TEST OF INDEPENDENCE (alpha = 0.05)

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Test A: Break frequency vs Eye discomfort level

eye_group Low (1-2) Moderate (3) High (4-5)

break_group

Few (0-3) 39 33 11

Moderate (4-6) 79 45 18

Frequent (7+) 56 15 4

chi2 = 13.403, dof = 4, p = 0.009467, Cramer's V = 0.149, min expected count = 8.2

Decision: Reject H0 -> significant association

Test B: Screen-time group vs Eye discomfort level

eye_group Low (1-2) Moderate (3) High (4-5)

screen_group

Low (<=4 h) 15 4 0

Medium (4-7 h) 93 40 17

High (>7 h) 66 49 16

chi2 = 8.691, dof = 4, p = 0.0693, Cramer's V = 0.120, min expected count = 2.1

Decision: Fail to reject H0 -> no significant association

Note: some expected counts < 5, interpret with caution.

Test C: Break frequency vs Concentration difficulty

conc_group Low (1-2) Moderate (3) High (4-5)

break_group

Few (0-3) 42 34 7

Moderate (4-6) 95 45 2

Frequent (7+) 66 7 2

chi2 = 30.100, dof = 4, p = 4.67e-06, Cramer's V = 0.224, min expected count = 2.8

Decision: Reject H0 -> significant association

Note: some expected counts < 5, interpret with caution.

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4. MULTIPLE LINEAR REGRESSION (OLS)

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Model 1: eye_discomfort ~ screen_hours + breaks_per_day

	coef	std err	t	p-value
intercept	2.0361	0.2206	9.2304	0.0000
screen_hours	0.1092	0.0271	4.0286	0.0001
breaks_per_day	-0.0836	0.0227	-3.6770	0.0003

n = 300, R2 = 0.091, Adj R2 = 0.085, F(2,297) = 14.83, p = 7.25e-07

Model 2: concentration_difficulty ~ screen_hours + breaks_per_day

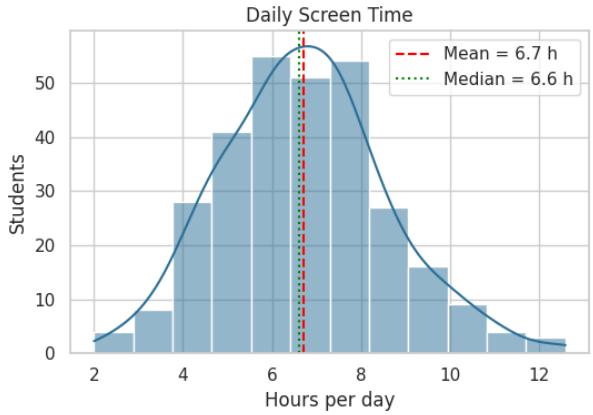
	coef	std err	t	p-value
intercept	2.1546	0.2048	10.5227	0.0000
screen_hours	0.0906	0.0252	3.6009	0.0004
breaks_per_day	-0.1361	0.0211	-6.4454	0.0000

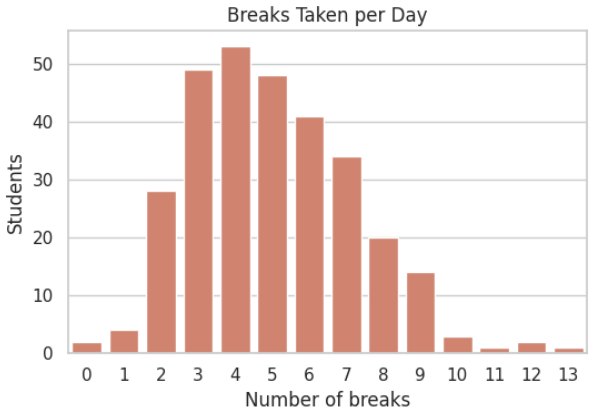
n = 300, R2 = 0.155, Adj R2 = 0.149, F(2,297) = 27.19, p = 1.44e-11

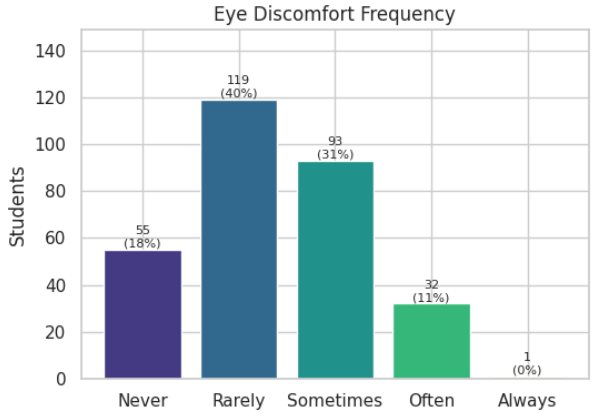
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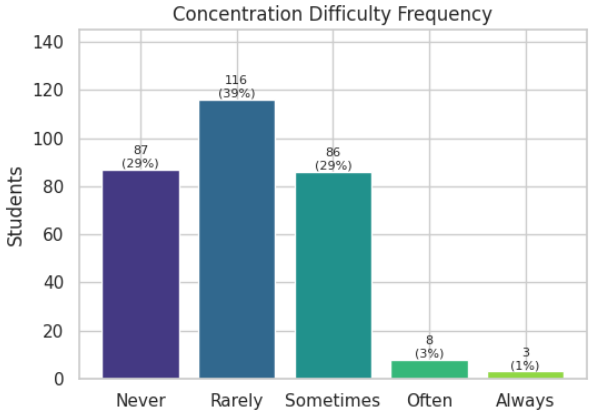
5. CHARTS

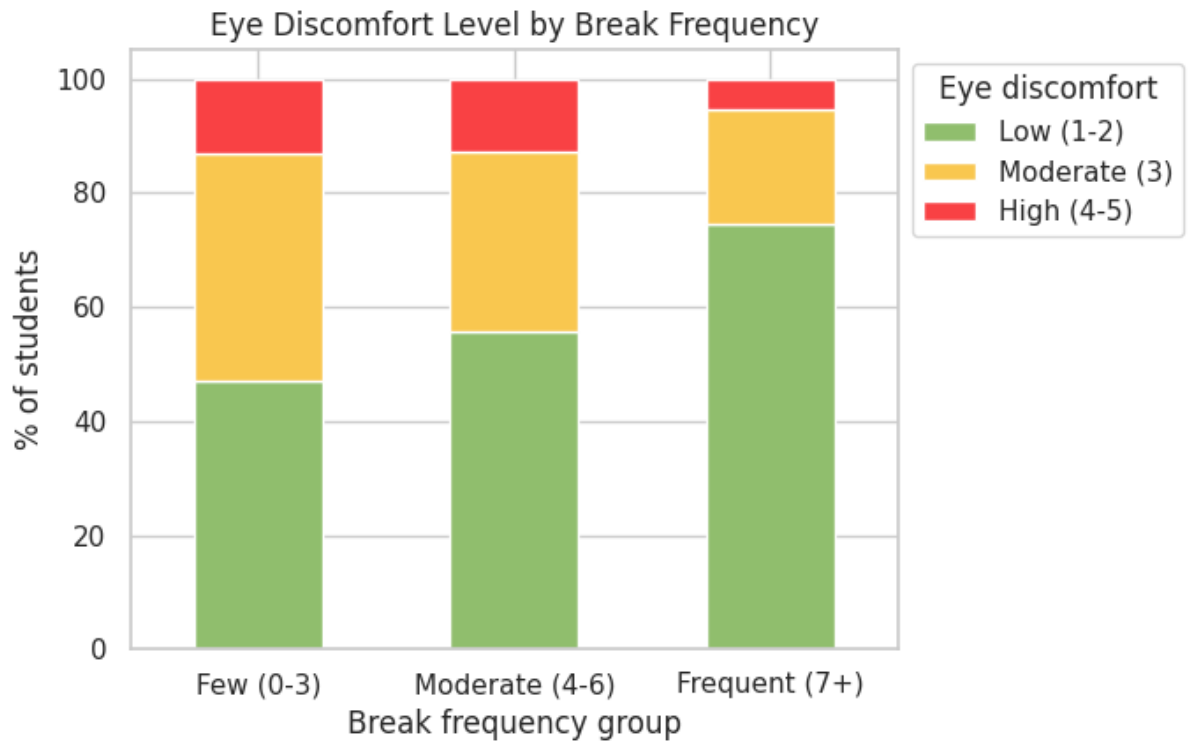
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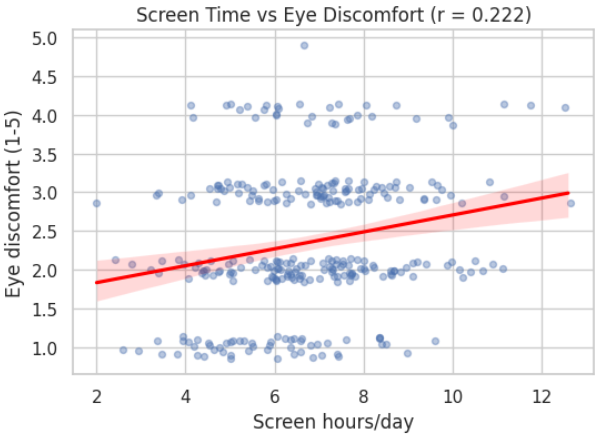

Daily Screen Time
Mean = 6.7 h
Median = 6.6 h
Students
Hours per day

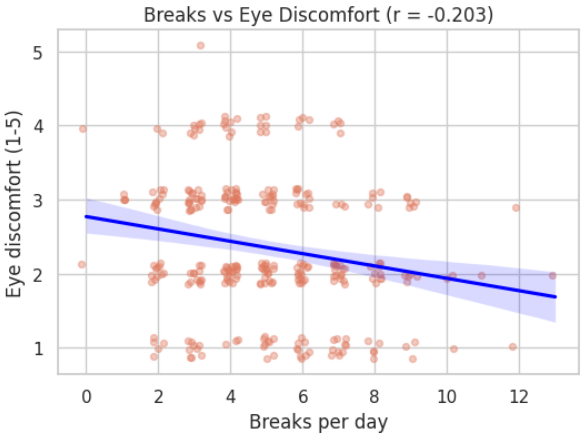

Breaks Taken per Day
Students
Number of breaks

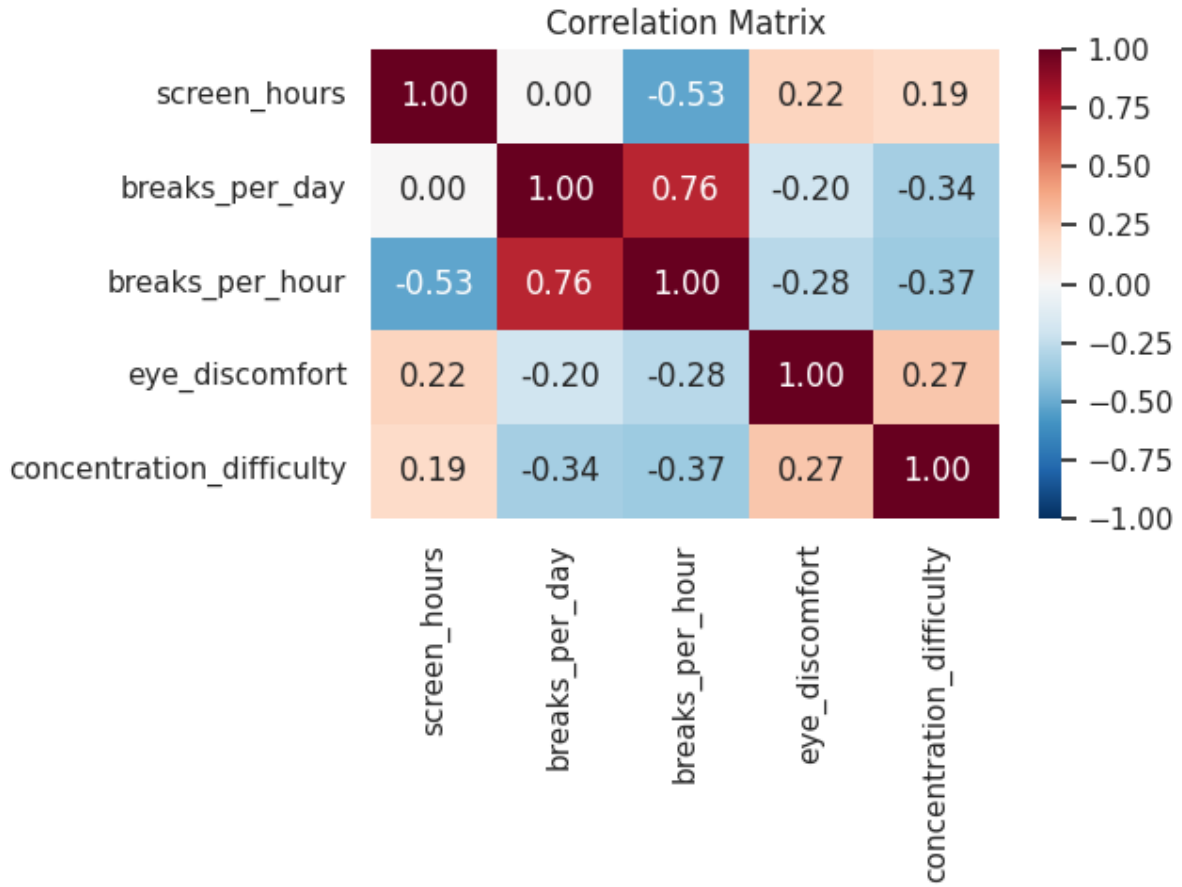

Eye Discomfort Frequency
Students
55 (18%)
119 (40%)
93 (31%)
32 (11%)
1 (0%)
Never
Rarely
Sometimes
Often
Always

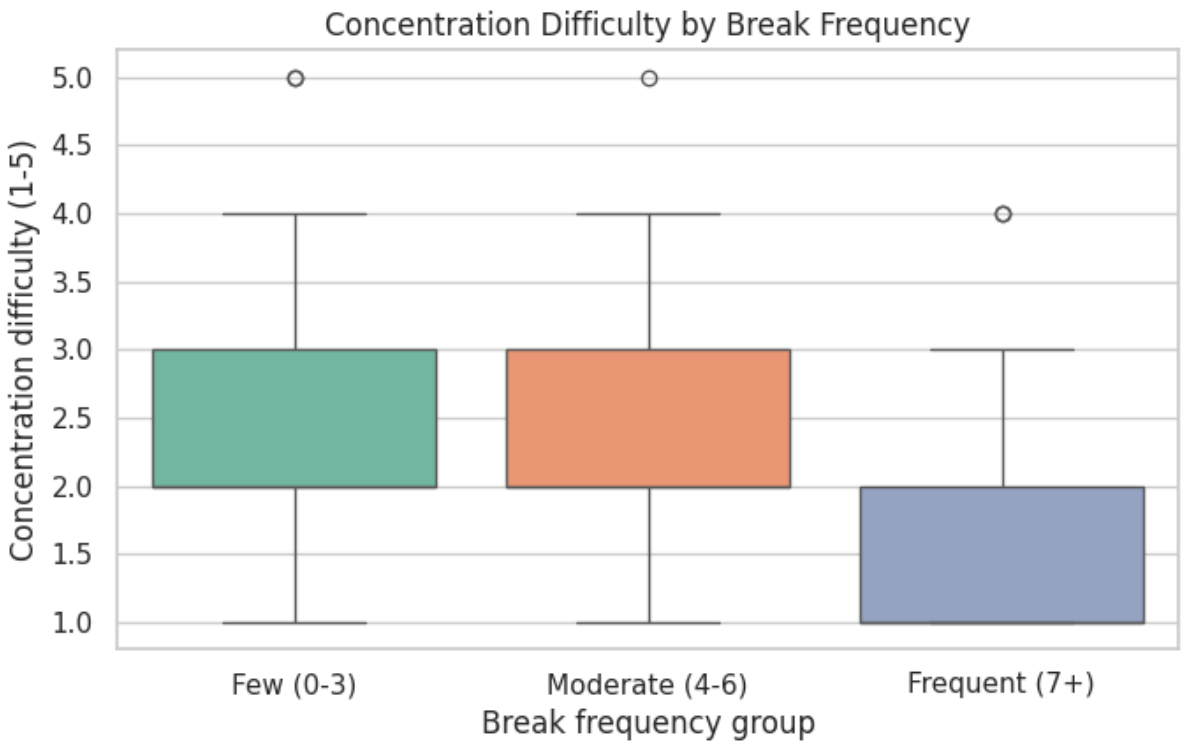

Concentration Difficulty Frequency
Students
87 (29%)
116 (39%)
86 (29%)
8 (3%)
3 (1%)
Never
Rarely
Sometimes
Often
Always

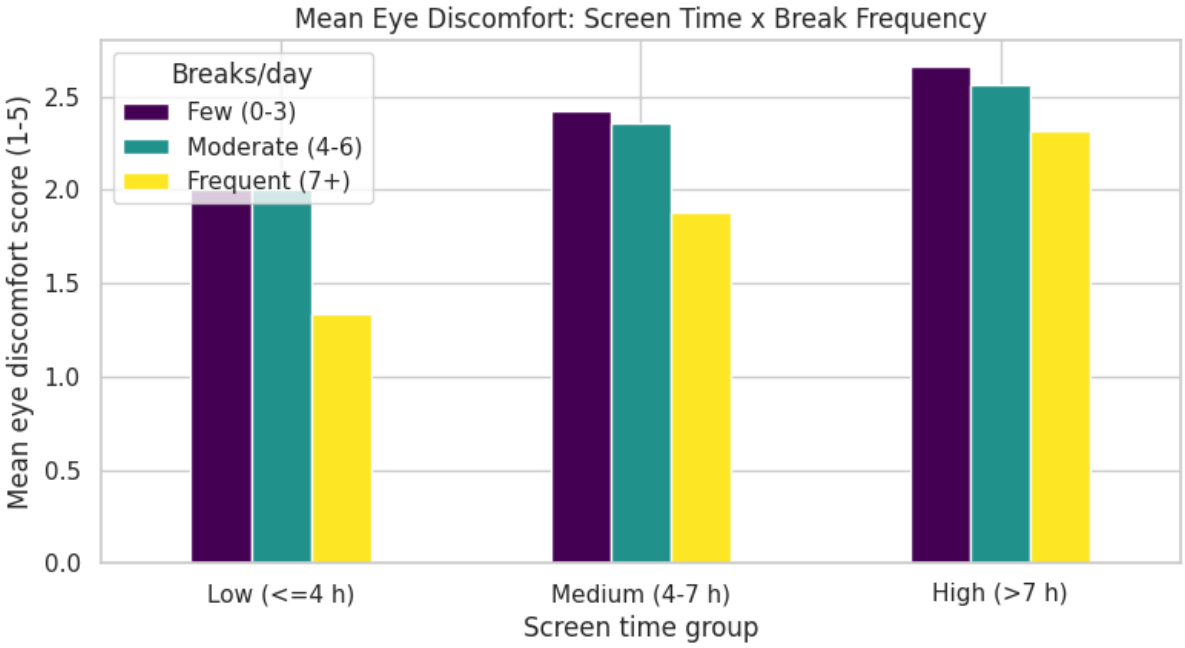

Eye Discomfort Level by Break Frequency
% of students
Few (0-3)
Moderate (4-6)
Frequent (7+)
Break frequency group
Eye discomfort
Low (1-2)
Moderate (3)
High (4-5)


Screen Time vs Eye Discomfort (r = 0.222)
Eye discomfort (1-5)
Screen hours/day


Breaks vs Eye Discomfort (r = -0.203)
Eye discomfort (1-5)
Breaks per day


Correlation Matrix
screen_hours
breaks_per_day
breaks_per_hour
eye_discomfort
concentration_difficulty
1.00 0.00 -0.53 0.22 0.19
0.00 1.00 0.76 -0.20 -0.34
-0.53 0.76 1.00 -0.28 -0.37
0.22 -0.20 -0.28 1.00 0.27
0.19 -0.34 -0.37 0.27 1.00





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6. AUTO-GENERATED SUMMARY

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* Sample size: 300. Mean screen time 6.72 h/day, mean breaks 5.02/day.

* Screen time vs eye discomfort r = 0.222; vs concentration difficulty r = 0.191.

* Breaks vs eye discomfort r = -0.203; vs concentration difficulty r = -0.343.

* Positive r = more of one goes with more of the other; negative r = more breaks go with fewer symptoms.

* Correlation and regression show association, not proof of cause.